Kite Games Presents Inter University Programming Contest - SUST CSE Carnival 2024

https://toph.co/c/inter-university-sust-cse-carnival-2024

Toph

Schedule

The contest will run for **5h0m0s**.

Authors

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Rules

This contest is formatted as per the official rules of ICPC Regional Programming Contests.

You can use C++17 GCC 13.2, C++20 Clang 16.0, C++20 GCC 13.2, Java 1.8, Kotlin 1.9, C GCC 5.3, C++11 GCC 5.3, C++11 GCC 7.4, C++14 GCC 5.3, C++14 GCC 8.3, C++98 GCC 5.3, C11 GCC 12.1, C11 GCC 9.2, Java, PyPy 7.1 (2.7), PyPy 7.1 (3.6), Python 2.7, Python 3.11, and Python 3.7 in this contest.

Be fair, be honest. Plagiarism will result in disqualification. Judges' decisions will be final.

Notes

There are 10 challenges in this contest.

Please make sure this booklet contains all of the pages.

If you find any discrepancies between the printed copy and the problem statements in Toph Arena, please rely on the latter.

A. Another Repeated String Problem

Let's define the $cost$ of a binary string p (consisting of only 0's and 1's) as the minimum positive integer k such that for every nonnegative integer $i < |p| - k$, $p_i = p_{i+k}$ holds.

Now, you are given a string s consisting of 0's and 1's and unknown characters '*'. Replace each unknown character '*' with either 0 or 1 in such a way that the string's $cost$ is minimized.

Print the minimum $cost$ of the string.

Input

The first line will contain T $(1 \leq T \leq 10)$, the number of test cases.

Each of the following T lines will contain the string s for that test case.

It's guaranteed that $\sum |s| \leq 10^6$.

Output

For each test case, print one integer, the minimum $cost$ of the string after modifying it.

Example

Input	Output
3 00*00**0 00*00**1 *0*1**1	1 5 3

B. Magic Grid

Let's define a $magic$ square grid of size $n \times n$ consisting positive integers as,

- If there is an integer x in the matrix, it must occur exactly x times.
- $\gcd(a_{ij},\ i) = 1$ and $\gcd(a_{ij},\ j) = 1$ for every i, j pair $(1 \leq i, j \leq n)$, where a_{ij} denotes the element at the i-th row and j-th column and $\gcd(x,\ y)$ denotes the greatest common divisor (GCD) of integers x and y.

You are given an integer n. Can you construct a $magic$ grid of size $n \times n$?

Input

The only line of the input contains a single integer n $(1 \leq n \leq 25)$.

Output

If there is no such $magic$ grid, print "-1" in a single line. Otherwise, print a $magic$ grid, i.e. n lines, i-th of which contains n space-separated integers, representing the i-th row of the grid.

If there are multiple answers, print any.

Examples

Input	Output
3	4 5 4 5 5 5 4 5 4

Input	Output
5	19 19 2 19 4 19 19 19 19 19 2 19 19 19 4 19 19 19 19 19 4 19 4 19 19

C. Rose Land

Roseland, the land of roses, can be represented as a complete binary tree with n nodes (cities) rooted at 1.

Alice and Bob live in the city u. Bob is a good friend of Alice. For higher study purposes, Alice is moving to another country. She has t days left before her flight. Bob wants to give her the maximum number of roses possible before her flight.

The i-th city has a rose garden that grows a_i roses per day. Bob can buy roses from city v which takes d days to deliver to their city u, where d is the distance between city u and city v i.e. the number of edges in the shortest path between the cities.

Now, you will be given q independent queries. Each query will contain two positive integers u and t where u is the city Alice and Bob live in (probably different for each query), and t is the number of days left before Alice leaves the country. Can you help Bob to calculate the maximum number of roses he can give Alice before she moves? For each independent query, let's assume that every garden has zero roses on day 0.

Note: A complete binary tree is a tree where each node, except the leaf nodes, has exactly two children, and all the leaves of the tree are at the same distance from the root.

Input

The first line will contain n $(1 \leq n \leq 2 \times 10^5)$ — the number of cities in Roseland.

The next line will contain n integers, $a_1, a_2, \ldots, a_n$ $(1 \leq a_i \leq 10^6)$ — the number of roses the i-th node can produce per day.

The following $n-1$ lines will contain two integers x and y $(1 \leq x,\ y \leq n)$ — denoting an edge between node x and y. It is guaranteed that the tree forms a valid complete binary tree.

The next line will contain an integer q $(1 \leq q \leq 10^5)$ — the number of queries.

The following q lines will contain two integers u and t $(1 \leq u \leq n,\ 1 \leq t \leq 10^6)$ — city they live in, days before Alice moves.

Output

Output q integers in q lines. The i-th integer is the answer for the i-th query.

Example

Input	Output
7 5 4 1 3 2 2 10 1 2 1 3 2 4 2 5 3 6 3 7 2 2 2 4 5	18 66

The first query result of the first sample test case is 18. In this query, city 2 can grow/produce $4 \times 2 = 8$ roses in 2 days. No transportation time is needed because Bob lives in the same city. City 1 can produce $5 \times 1 = 5$ roses in 1 day and transport them to the city 2 in 1 day. City 4 can produce $3 \times 1 = 3$ roses in 1 day and send them to the city 2 in 1 day. Similarly, city 5 will send $2 \times 1 = 2$ roses. City 3, 6, and 7 can't send any roses to city 2 because it takes at least 2 days just to transport any roses from them, therefore is no time for producing roses. So in total, Bob can arrange a maximum of $8 + 5 + 2 + 3 = 18$ roses for Alice before she moves.

D. Connecting Dots

You are given n distinct points $p_1,\ p_2,\ p_3,\ \ldots\ ,\ p_n$. Point p_i has coordinates $(x_i,\ y_i)$ in the 2D plane.

You can pick any two points $p_i(x_i,\ y_i)$ and $p_j(x_j,\ y_j)$, and draw a segment between them, if and only if these three conditions are met:

1. $x_i\ /\ x_j =$
2. There is no such k for which $x_i < x_k < x_j$
3. This segment doesn't intersect with any other previously drawn segments (but can touch at the endpoint).

Your task is to determine the maximum number of segments you can draw.

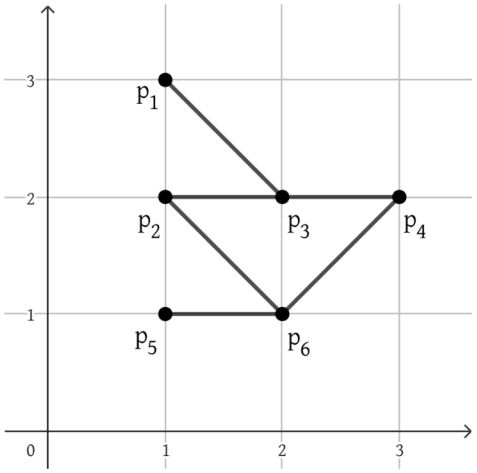


Fig: 1

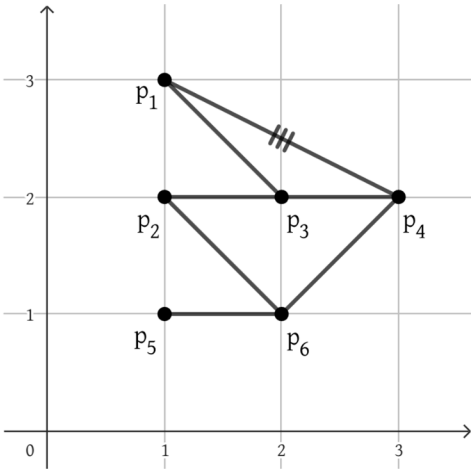


Fig: 2

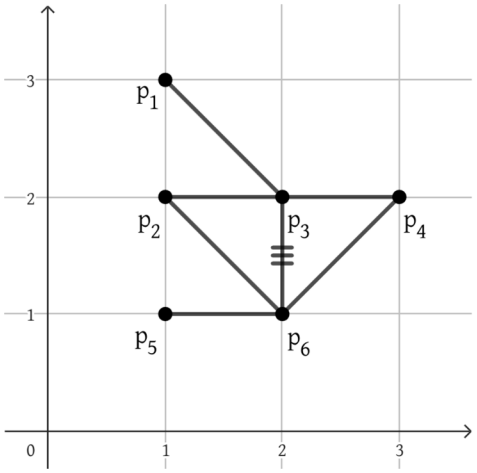


Fig: 3

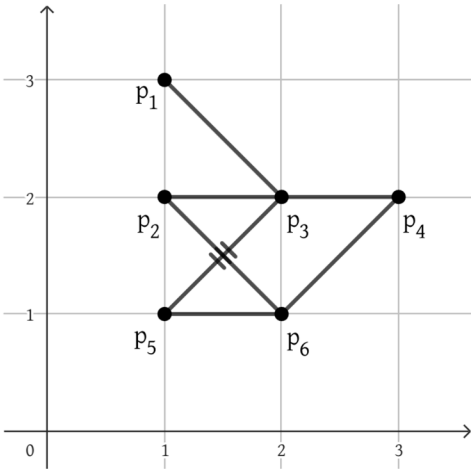


Fig: 4

Figures above describe the points from the first sample test case. Fig 1 is the correct answer, whereas Fig 2, 3, and 4 are invalid. Fig 2 violates rule 2, Fig 3 violates rule 1, and Fig 4 violates rule 3.

Input

The first line of the input will contain a single integer T $(1 \leq T \leq 100)$ — the number of test cases.

For each test case, the first line will contain n $(1 \leq n \leq 100)$ — the number of points.

The following n lines of each test case will contain two integers $x_i,\ y_i$ $(1 \leq x_i,\ y_i \leq 100)$ — denote the coordinates for the i-th point.

Output

For each test case, print one single integer — the maximum number of segments you can draw for that test case.

Example

Input	Output
2 6 1 3 1 2 2 2 3 2 1 1 2 1 4 1 1 1 2 2 1 2 2	6 3

E. Hottest Problem of The Winter

Dear Contestants,

Welcome to "Kite Games Presents Inter University Programming Contest - SUST CSE Carnival 2024". Isn't it windy today? Let's warm up a little bit with the hottest problem of this winter. Do you know what a camel case is? According to Mozilla documentation, "Camel case is a way of writing phrases without spaces, where the first letter of each word is capitalized, except for the first letter of the entire compound word, which may be either upper or lower case".

For example, if we choose these three words, ["Toph", "is", "great"], the proper camel case representation would be TophIsGreat, or tophIsGreat. But tophisgreat or tophISGReat are not valid.

So given a list of words only containing Latin letters, can you generate a camel case representation of it?

Input

The first line will contain n $(1 \leq n \leq 100)$, the number of words.

The next line will contain n words separated by spaces, we can assume that no word is more than 10 characters long.

Output

Output only one single string, the camel case representation of given words.

Example

Input	Output
3 Kite GaMes Studio	KiteGamesStudio

F. Infinite Tree

Given an infinite tree (forest?), find the distance between two nodes for some queries. You know, it's tough to describe infinite things. So we kinda hacked it out for you, so we give you an array of positive integers a of length m (0-indexed). And we say the n-th node is connected to $n - a_{n \pmod m}$. Because it is an infinite tree, the node numbers can be negative as well.

Note: $x \pmod m$ is equal to y, if and only if $0 \leq y < m$ and m divides $x - y$.

Input

The first line will contain two integers m and q $(1 \leq m,\ q \leq 10^5)$ and the following line will have m positive integers which describe the array a $(1 \leq a_i \leq 10^5)$.

Following q lines will contain two integers u_i and v_i $(-10^9 \leq u_i,\ v_i \leq 10^9)$ — two nodes of the tree.

Output

For each query, print an integer, the distance between the nodes u_i and v_i or **-1** if they are disconnected.

Example

Input	Output
3 3 1 1 1 6 8 5 100 -3 18	2 95 21

G. Half Empty or Half Full?

There are N glasses in a row numbered from 1 to N on a table. The i-th glass has a volume of V_i liters and has W_i liters of water in it.

A glass is considered "good" if the amount of water is **more than** half of its volume, otherwise, it is a "bad" glass. As a skilled waterbender and earthbender, you can choose K **non-intersecting** segments of glasses, and use your bending skills for each segment to merge them into a new glass that has the same amount of volume and water as the sum of volumes and waters of that segment. Formally, if you merge glasses in segment l r, you get a new glass with volume $V = \sum_{i=l}^{r} V_i$ and water $W = \sum_{i=l}^{r} W_i$.

Now by choosing any such K segments and merging them with your bending skill, is it possible to have **more good glasses than bad**?

Note that, two segments (l, r) and (p, q) are non-intersecting when there's no common index j such that $l \leq j \leq r$ and $p \leq j \leq q$.

Input

The first line of input contains T $(1 \leq T \leq 100)$, the number of test cases.

The first line of each test case contains an integer N $(1 \leq N \leq 10^5)$, the number of glasses on the table initially.

The second line of each test case contains N integers $V_1, V_2, \ldots, V_N$ $(1 \leq V_i \leq 10^9)$, where V_i represents the volume of the i-th glass. The third line of the test case contains another N integers $W_1, W_2, \ldots, W_N$ $(0 \leq W_i \leq V_i \leq 10^9)$, where W_i represents the amount of water in the i-th glass.

The sum of N over all test cases does not exceed 2×10^5.

Output

For each test case, if it is possible to have more good glasses than bad, print "Yes" and in the next line print K $(0 \leq K \leq N)$. For the following K lines, print l_i, r_i $(1 \leq l_i < r_i \leq N)$, representing the non-intersecting segments of glasses you merged together. If there are multiple possible ways, you can output any. Print "No" otherwise.

Example

Input	Output
2 4 6 4 8 10 4 1 6 1 3 4 4 5 1 2 3	Yes 1 2 3 No

H. Problem Title Will Be Provided Later

Consider a list of integers, denoting a, of length n. The task is to find the maximum value of $F(l,\ r)$, where l and r represent indices within the list and $l \leq r$. The definition of $F(l,\ r)$ is given below.

$$F(l,\ r) = (\max_{l \leq i \leq r} a_i) \times (\sum_{j=l}^{r} a_j)$$

Input

The input begins with a single positive integer T $(1 \leq T \leq 10^5)$ — the number of test cases.

Subsequently, each of the following T test cases starts with a positive integer n $(1 \leq n \leq 10^5)$ — the number of elements within the list denoted as a.

Following this, a line containing n space-separated integers is provided to specify the values of the elements in the list. Each a_i will maintain the constraint $-10^5 \leq a_i \leq 10^5$.

Note that the sum of n over all test cases will not exceed 10^6.

Output

For each test case, print the answer in a new line.

Example

Input	Output
2 4 1 2 3 4 4 1 2 3 -4	40 18

I. Cut The Polygon

Given a convex polygon of n vertices, select three distinct vertices a, b, c, and draw line segments ab and ac that divide the polygon into three pieces (of possibly zero areas). Let the areas of the polygon be A and the area of the three pieces be A_1, A_2, and A_3.

Find the minimum value of $|A_1 - \frac{A}{3}| + |A_2 - \frac{A}{3}| + |A_3 - \frac{A}{3}|$ by selecting a, b, and c optimally.

Input

The first line contains a single integer n $(3 \leq n \leq 10^5)$ — the number of vertices of the polygon.

The following n lines describe vertices of the polygon in the counterclockwise direction. The i-th line contains two integers $x_i,\ y_i$ $(|x_i|, |y_i| \leq 10^8)$ — the coordinates of the i-th vertex.

It is guaranteed that the polygon is convex and that no two adjacent sides are parallel.

Output

Print the minimum value of the expression **multiplied by 6**. It can be proven that it will always be an integer.

Examples

Input	Output
3 0 0 2 0 0 2	16

Input	Output
5 0 0 5 0 4 2 2 4 0 5	8

J. No Expectation

Blessed is he who expects nothing, for he shall never be disappointed.

It is expectations that give birth to all disappointments. And despite that, as a competitive programmer, you must expect, i.e. compute expected values.

You have n metaphorical sticks of lengths $a_1, a_2, \ldots, a_n$. Using these sticks, you will perform a particular operation k times. In one operation, you will choose one of the existing sticks uniformly at random and then the chosen stick will be broken into two parts of equal length. So each operation will increase the number of sticks by 1. After these operations, you are interested in the sum of all pairwise products of the stick lengths. To be more formal, if the stick lengths after k operations are $s_1, s_2, \ldots, s_{n+k}$ then the value you are interested in is $\sum_{i<j} s_i s_j$. Your job is to find out the expected value of this expression. Since computers are not famous for handling floating-point calculations, you have to compute this expected value modulo 998244353. So if the answer is a fraction $\frac{p}{q}$ you have to output $p \times q^{-1} \bmod 998244353$ where q^{-1} is the multiplicative inverse of $q \bmod 998244353$.

Input

The first line of the input will contain a positive integer T $(1 \leq T \leq 10^5)$, the number of independent test cases. T test cases will follow.

First line of each test case will contain two positive space-separated integers n and k $(1 \leq n,\ k \leq 10^5)$.

Next line will contain the n space separated positive integers $a_1, a_2, \ldots, a_n$ $(1 \leq a_i \leq 10^8)$.

Note that, $\sum n \leq 10^6$ and $\sum k \leq 10^6$.

Output

Output T lines. For each test case output the expected value modulo 998244353 in a separate line.

Example

Input	Output
2 3 1 4 2 2	22 748683288

Input	Output
3 2 4 2 2	

In the first test case, after the first operation, you have the following stick lengths possible: $[2,\ 2,\ 2,\ 2], [4,\ 1,\ 1,\ 2], [4,\ 2,\ 1,\ 1]$. Their pairwise product sums are $24,\ 21,\ 21$, the expected value is 22.

In second testcase the possible stick length sets are: $[2,\ 2,\ 2,\ 1,\ 1]$ in 4 ways, $[2,\ 2,\ 2,\ 1,\ 1]$ in 2 ways, $[4,\ 1,\ 1,\ 1,\ 1]$ in 2 ways, $[4,\ 2,\ 1,\ \frac{1}{2},\ \frac{1}{2}]$ in 4 ways.